

2. [Maximum mark: 7]

Bob invests 1000 dinar in an account which pays a nominal annual interest rate of 4% compounded **quarterly**.

The amount of money in the account after one complete year can be written as $1000(1 + k)^4$ where $k \in \mathbb{Q}$.

- (a) Write down the value of k . [1]
- (b) Expand and simplify $(1 + x)^4$. [2]
- (c) Hence or otherwise, find the amount of money in the account after one complete year, giving your answer correct to the nearest dinar. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

